# Fitting Linear Regression Models: Feature Engineering, Feature Selection, and Model Evaluation

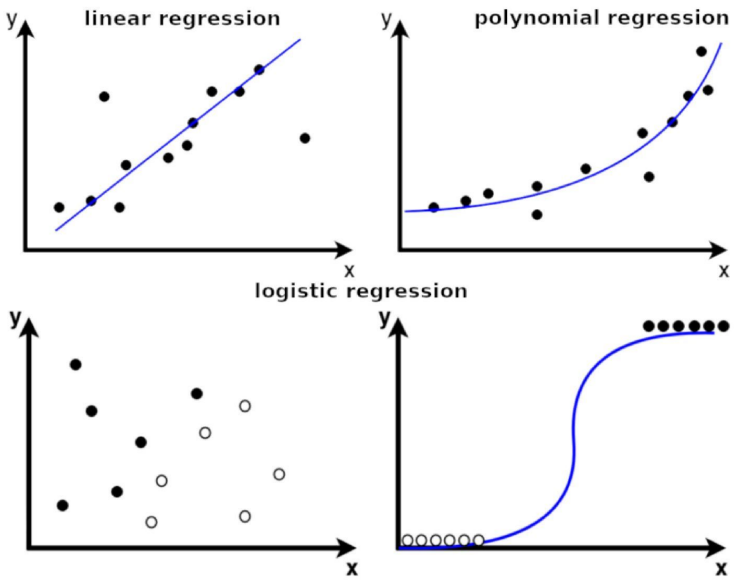
DATA 201 -- WEEK 7

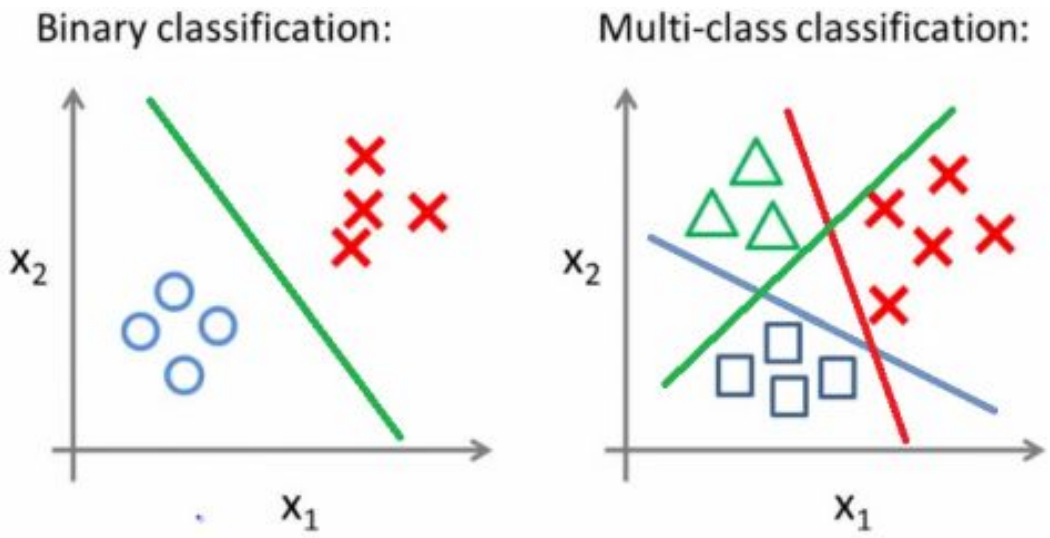
SPRING

# Week 5 Learning Objectives

- Distinguish **model form decisions** from **feature inclusion decisions** for fitting linear regression models
- **Part 1 - Model Form Decisions (Feature Engineering)**
  - Use **EDA and residual plots** to diagnose problems in linear regression
  - Identify when **transformations** of variables may be needed
  - Identify when **interaction terms** may be needed
  - Use **AIC/BIC** to evaluate whether diagnostic-driven changes improve the model
- **Part 2 – Feature Inclusion Decisions (Importance/Selection)**
  - Apply feature-selection techniques specific to linear regression
  - Use adjusted $R$2R2 and AIC/BIC to compare predictor sets
  - Diagnose and manage multicollinearity
  - Justify feature inclusion using statistical evidence + domain reasoning

# Background and Context Recap



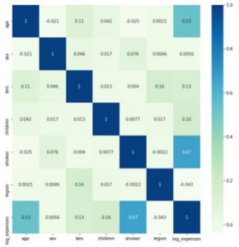

# Regression in Context: Statistical Learning

# LINEAR REGRESSION

## 1 EDA

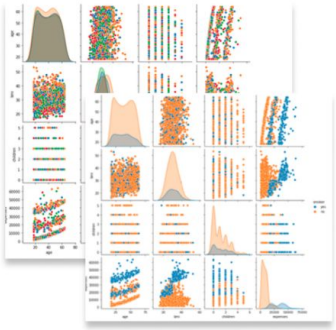
### Heatmap

*sns.heatmap(df.corr( ))*



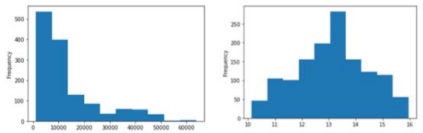
### Scatterplot

*sns.pairplot(df, hue = "...")*



## 2 Feature Engineering

### Log Transformation *np.log2( )*



### One Hot Encoder

*pd.get_dummies( )*

| smoker_no | smoker_yes | region_northeast | region_northwest | region_southea |
|---|---|---|---|---|
| 0 | 1 | 0 | 0 | |
| 1 | 0 | 0 | 0 | |
| 1 | 0 | 0 | 0 | |
| 1 | 0 | 0 | 1 | |
| 1 | 0 | 0 | 1 | |

### Label Encoder

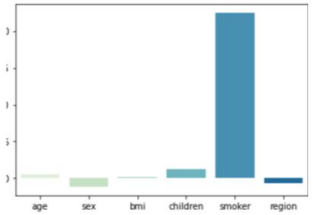
*LabelEncoder( )*

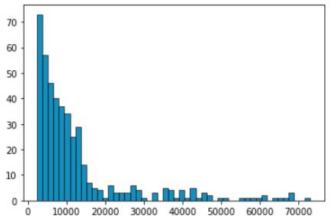
| sex | bmi | children | smoker | region |
|---|---|---|---|---|
| 0 | 27.9 | 0 | 1 | 3 |
| 1 | 33.8 | 1 | 0 | 2 |
| 1 | 33.0 | 3 | 0 | 2 |
| 1 | 22.7 | 0 | 0 | 1 |
| 1 | 28.9 | 0 | 0 | 1 |

## 3 Implementation

### Feature Importance

*sns.barplot(x = X_train.columns, y = coef)*



### Linear Regerssion

*model = LinearRegression( )*
*model.fit(X_train, y_train)*
*y_pred = model.predict(X_test)*
*coef = model.coef_*
*intercept = model.intercept_*

## 4 Evaluation

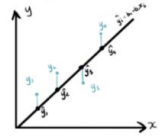
### R Squared

coefficient of determination

$$R^2 = 1 - \frac{RSS}{TSS}$$

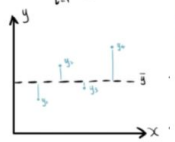
Residual Sum of Squares (RSS)

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$



Total Sum of Squares (TSS)

$$TSS = \sum_{i=1}^{n} (y_i - \bar{y})^2$$



### Mean Absolute Error

$$MAE = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i|$$

### Mean Squared Error

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

### Root of Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2}$$

visit www.visual-design.net for step by step guide

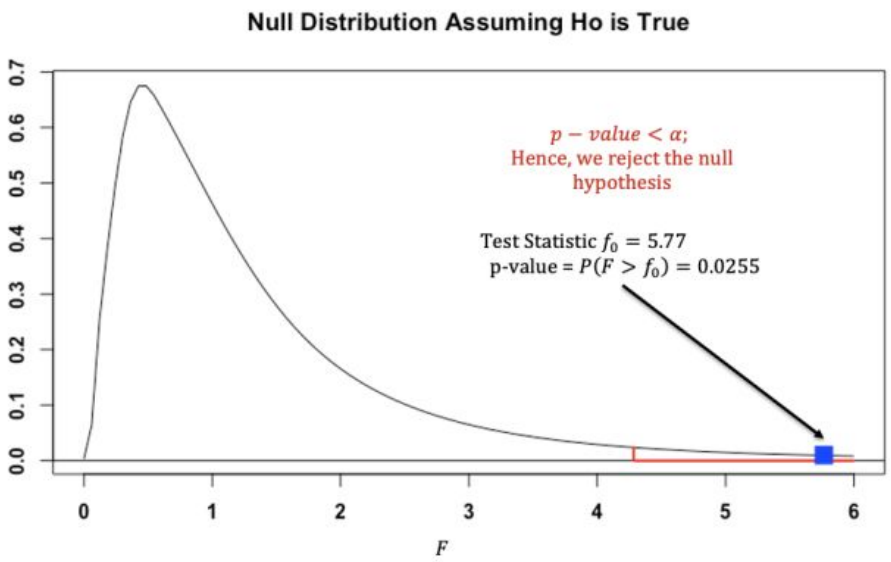
# Basic Evaluation and Diagnostics for Multiple Linear Regression

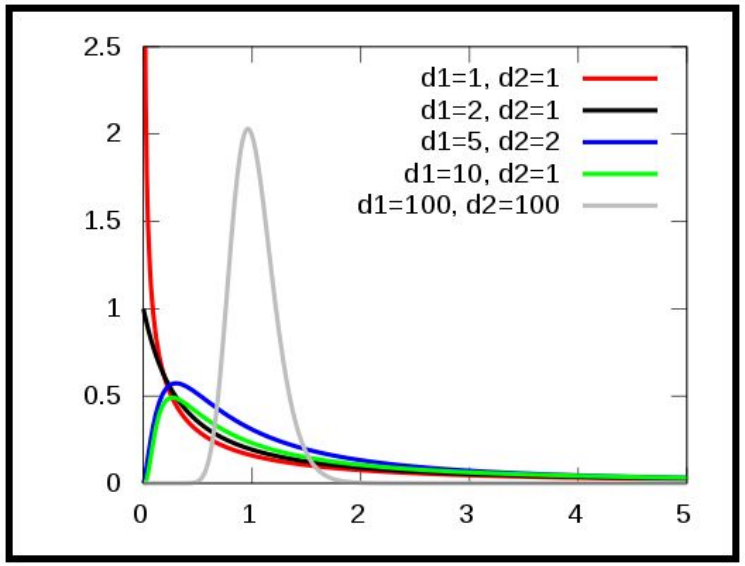
- **Model Fit:** These tell you how well the regression equation explains the variability in your dependent variable.
  - **$R^2$ (Coefficient of Determination)**
  - **Adjusted $R^2$**
  - **F-statistic (overall model significance)**
- **Coefficient-Level Significance**
  - **t-tests for each coefficient**
  - **Confidence Intervals (95%)** for coefficients
- **Assumption Diagnostics** for multiple linear regression techniques to be valid
  - Linearity
  - Independence
  - Homoscedasticity (constant variance)
  - Errors are normally distributed
  - No multicollinearity
- **Residual Analysis** – residuals should be normally distributed

# Key Questions for Multiple Regression (1)

- Is there a relationship between response and predictors (that can be captured in a linear model?)

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_p = 0 \qquad H_a : \text{ at least one } \beta_j \text{ is non-zero.}$$



# Recap: Linear Regresson
# Estimation (Inference) vs Prediction

**Estimation**: What is the mean weight, $\mu$, of **all** American women aged 18-24?

- We can calculate a confidence interval for $\mu$ using $\bar{y}$
- **Confidence interval** for the mean response when the predictor value is $x$

**Prediction:** What is the weight, $y$, of an **individual** American woman, aged 18-24?

- We can calculate a prediction interval for $y$ using $\bar{y}$
- **Prediction interval** for the individual response when the predictor value is $x$

# Recap: Linear Regression Prediction and Testing (Inference)

## Prediction

Linear Regression: Single Variable

$$\boxed{\hat{y}} = \underbrace{\beta_0 + \beta_1}_{\text{Coefficients}} \boxed{x} + \boxed{\epsilon}$$

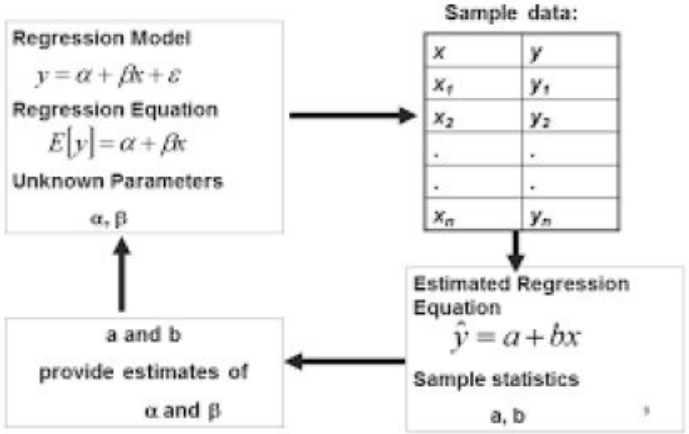
Predicted output — Coefficients — Input — Error

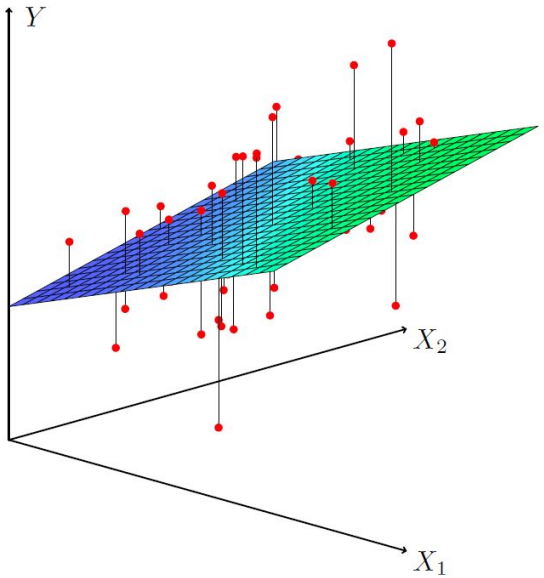
Linear Regression: Multiple Variables

$$\boxed{\hat{y}} = \underbrace{\beta_0 + \beta_1}\boxed{x_1} + \ldots + \underbrace{\beta_p}\boxed{x_p} + \boxed{\epsilon}$$

## Inference:

## Null hypothesis slope = 0






Estimate parameters that minimize the sum of squared residuals (RSS)

Dependent Variable (Response Variable)

Independent Variables (Predictors)

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \varepsilon$$

Y intercept

Slope Coefficient

Error Term

- $\beta_i$ is the *average effect* on Y of one unit increase in $X_i$, ***holding all other predictors fixed***.
- Ideally, predictors are independent and uncorrelated
- Each coefficient can be estimated and tested separately

- ***Reality:*** **predictors usually change together**
- **Unless there is an experimental design (random assignment of treatments), *not possible to infer causality***

# Fit Model = Minimize the Loss Function

A loss function quantifies the error between the observed (target) value and the predicted value

## TRADITIONAL: ORDINARY LEAST SQUARES/MEAN SQUARED ERROR

- Find the error, $e_i = y_i - \hat{y}_i$
- Take the square of the error (*also known as the residual sum of squares or RSS*)

$e_i{}^2 = (y_i - \hat{y}_i)^2$

- Take the mean of the squared errors

$$\text{MSE} = \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n} = \frac{\sum_{i=1}^{n} e_i{}^2}{n}$$

- Preferred when penalizing significant errors is crucial
- Tends to yield more smooth and continuous fits

## COMMON ALTERNATE: MEAN ABSOLUTE ERROR

- Find the error, $e_i = y_i - \hat{y}_i$
- Take the absolute value of the error $|e_i| = |y_i - \hat{y}_i|$
- Take the mean of the absolute value of the errors

$$\text{MAE} = \frac{\sum_{i=1}^{n}|y_i - \hat{y}_i|}{n} = \frac{\sum_{i=1}^{n}|e_i|}{n}$$

- Less sensitive to outliers
- Consistent error measurement across all data points

# Part 1 - Model Form Decisions: Feature Engineering

# Main Objectives

- Use **EDA and residual plots** to diagnose problems in linear regression
- Identify when **transformations** of variables may be needed
- Identify when **interaction terms** may be needed
- Use **AIC/BIC** to evaluate whether diagnostic-driven changes improve the model (See Part 3: Model Evaluation)

# Recall: Assumptions for Modeling with Linear Regression

- Linearity
- Independence
- Homoscedasticity (constant variance)
- Errors are normally distributed
- No multicollinearity

- We use EDA and diagnostics (residuals) to identify when these assumptions aren't met and introduce modifications to features that may address the

# When Assumptions are Violated (1)

- **Linearity Violation Problem**: The relationship between predictors and the outcome isn't linear.
  - **Transform variables** (e.g., log, square root, polynomial terms).→ Example: use np.log(X) or X**2 in Python.
  - Add **interaction terms** if variables jointly affect the outcome.
  - **Use nonlinear models** like decision trees or generalized additive models (GAMs).
- **Independence Violation:** Residuals are correlated — often in time series or spatial data.
  - Include **lag variables** to capture temporal effects.
  - Use **time-series models** such as ARIMA or GLS (Generalized Least Squares).
  - **Cluster or group data** if observations are dependent within groups (use mixed-effects models).

# When Assumptions are Violated (2)

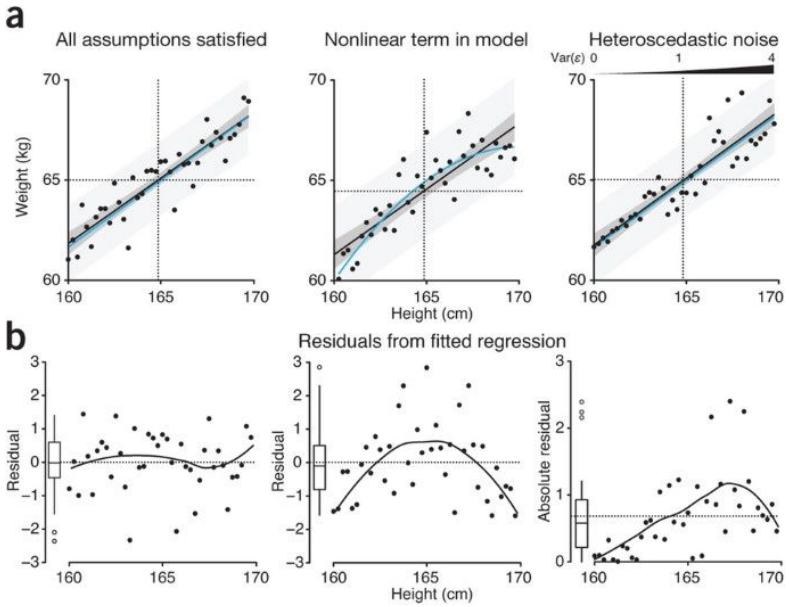
- **Homoscedasticity Violation (Unequal Variance)Problem**: The spread of residuals increases or decreases with predicted values.
  - **Transform** the dependent variable (e.g., log(Y), sqrt(Y)) to stabilize variance.
  - Use **weighted least squares** (WLS) to give less weight to points with high variance.
  - Switch to **robust regression methods** (e.g., Huber or RANSAC), which are less sensitive to heteroscedasticity.

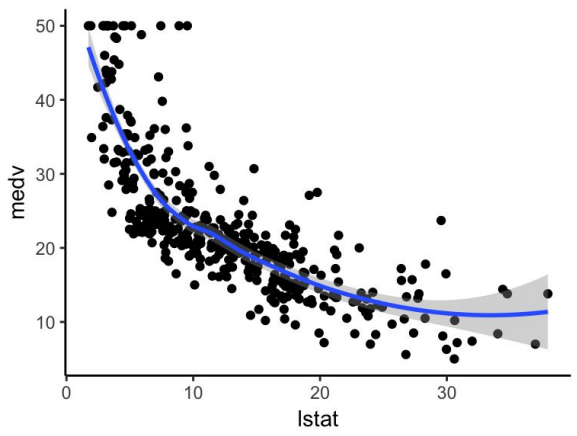
- **Normality Violation of Residuals Problem**: Residuals aren't normally distributed — affects hypothesis testing and confidence intervals.
  - **Transform Y** (e.g., log, Box-Cox transformation) to achieve normality.
  - Rely on **large-sample theory**: With large datasets, normality matters less due to the Central Limit Theorem.
  - Use **nonparametric or bootstrapped confidence intervals** instead of relying on normal-based tests.

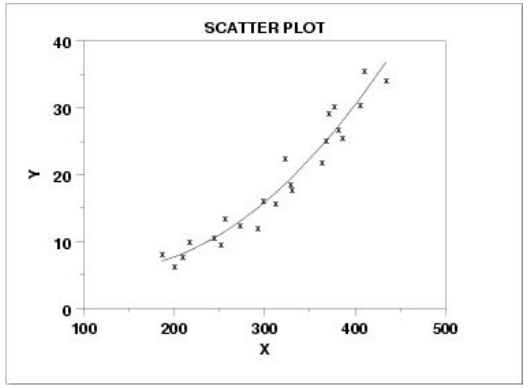
# When Assumptions are Violated (3)

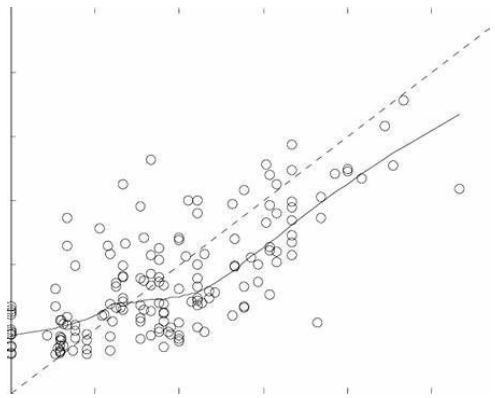
- **Outliers and Influential Points** Some data points strongly affect the regression line.
  - **Check Cook's distance or leverage values** to identify them.
  - **Remove or winsorize** (cap extreme values) after verifying data entry is correct.
  - **Use robust regression** to reduce sensitivity to outliers.

  - To be discussed in Part 2





# Objective 1: Use EDA and residual plots to diagnose problems in linear regression (1)

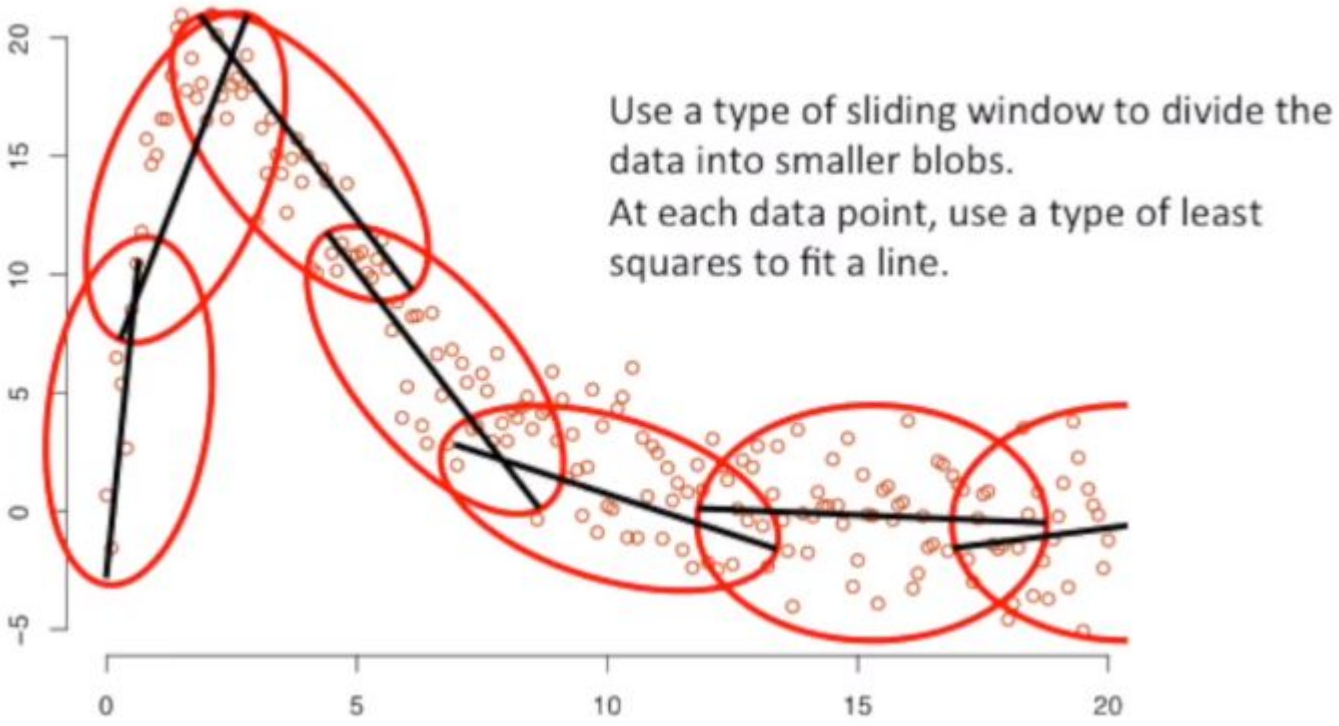
- **Correlation Heatmaps** (Baseline features)
- **Scatterplots with Smooths** (Pre-Model)
  - Scatterplots of $Y$ vs. Predictor
  - May add a loess smoother for visual guidance
- **What to look for**
  - Systematic curvature ⟶ missing nonlinear term
  - Sharp early slope then flattening ⟶ consider log(X)
  - Fan-shaped spread ⟶ consider transformation of Y

# Local Fitting for Simple Linear Models

- **Running Mean or Median** – Take a small interval around x. and estimate f(x) based on the mean (or median) of the y-value of points in that interval.
- **Kernel Smoothing** – uses the weighted average of nearby data points to estimate the value of the function
- **LOESS (locally estimated scatterplot smoothing)** - "a nonparametric method for smoothing a series of data in which no assumptions are made about the underlying structure of the data. LOESS uses local regression to fit a smooth curve through a scatterplot of data. " (Cleveland, W.S., 1979)

Source: http://scott.fortmann-roe.com/docs/BiasVariance.html

# LOESS Regression



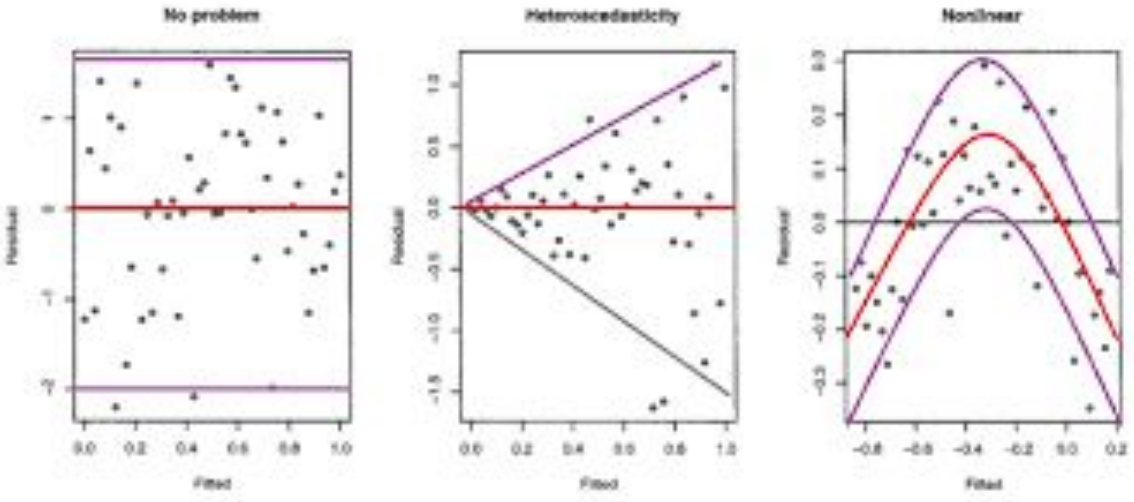

**Pros**:

- Simple and flexible
- No assumptions about relationship
- Valuable for visualization
- New values can be estimated for validation

**Cons**:

- Requires densely sampled data
- No formula produced – difficult to transfer or replicate
- Computationally intensive
- Sensitive to outliers

# Objective 1: Use EDA and residual plots to diagnose problems in linear regression (2)




- **Residuals vs Fitted (Post Model)**
  - Fit Y ~ X (+ other terms)
  - Plot residuals vs X
  - Curved residual trend
    - Functional form misspecified → add polynomial/transform
  - After fix: residuals should be patternless around 0

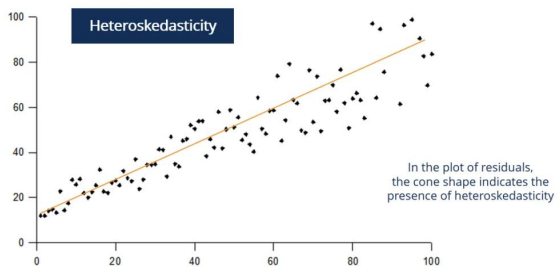
| Residual pattern | Interpretation | Action |
|---|---|---|
| Curved trend | Functional form misspecified | Add polynomial or transform |
| Systematic slope | Missing interaction or nonlinear term | Investigate conditioning |
| Increasing spread | Non-constant variance | Transform Y |

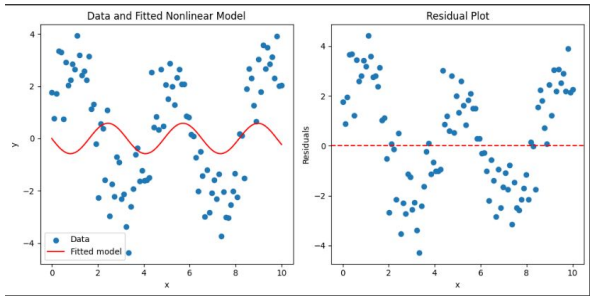
# Objective 2: Identify when transformations of variables may be needed (1)

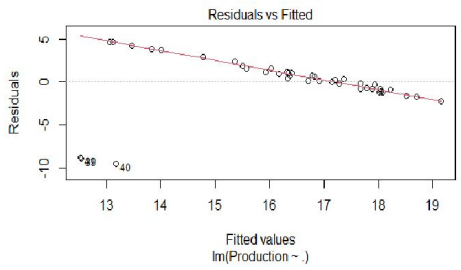
- Examine **skewness and variance** in histograms and boxplots

| Pattern | Suggestion |
|---|---|
| Right-skewed X | log(X) |
| Right-skewed Y | log(Y) |
| Multiplicative relationship | log–log model |

# Objective 2: Identify when transformations of variables may be needed (2)



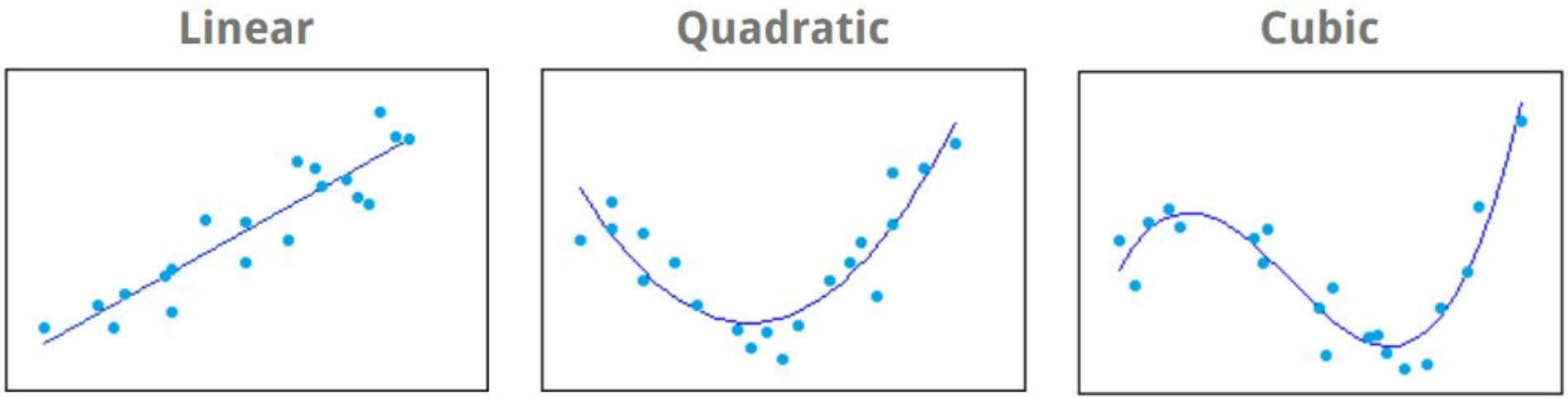




**Advanced interpretation of residuals**

- Funnel shape ⟶ variance stabilization needed
- Curvature ⟶ wrong functional form
- Clusters ⟶ missing grouping variable or interaction

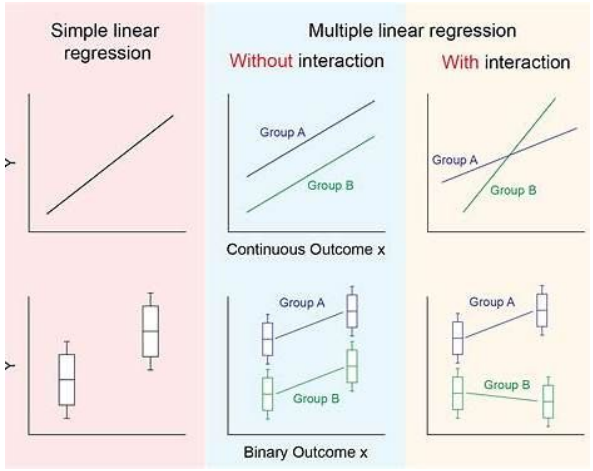
# Transformations - Polynomial

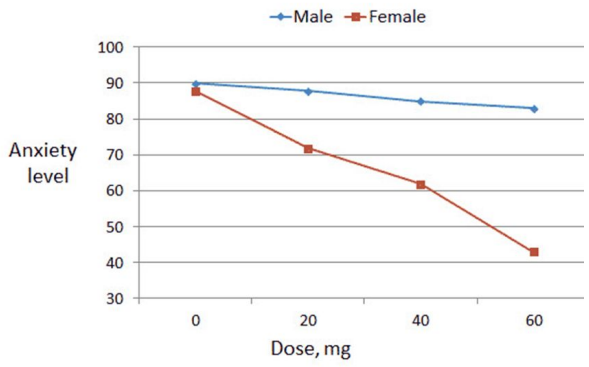


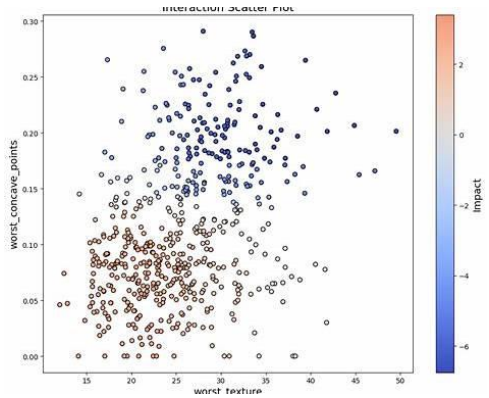

$$Y \sim a + bX + bX^2 + bX^3 + \ldots + bX^n + \epsilon$$

| Case | Equation | $\beta$ | Interpretation |
|---|---|---|---|
| 1 | $Y = \alpha + \beta X$ | $\frac{\Delta Y}{\Delta X}$ | Unit change of Y in response to a unit change of X |
| 2 | $\log(Y) = \alpha + \beta \log(X)$ | $\frac{\Delta \log Y}{\Delta \log X} = \frac{100 \Delta \log Y}{100 \Delta \log X}$ | % change of Y in response to 1% change of X (elasticity) |
| 3 | $Y = \alpha + \beta \log(X)$ | $\frac{\Delta Y}{\Delta \log X} = \frac{100 \Delta Y}{100 \Delta \log X}$ | (β/100) unit change of Y in response to 1% change of X |
| 4 | $\log(Y) = \alpha + \beta X$ | $\frac{\Delta \log Y}{\Delta X}$ | 100β% change of Y in response to 1 unit change of X |

# Other Transformations: Log-linear




# Objective 3: Identify when interaction terms may be needed (1)

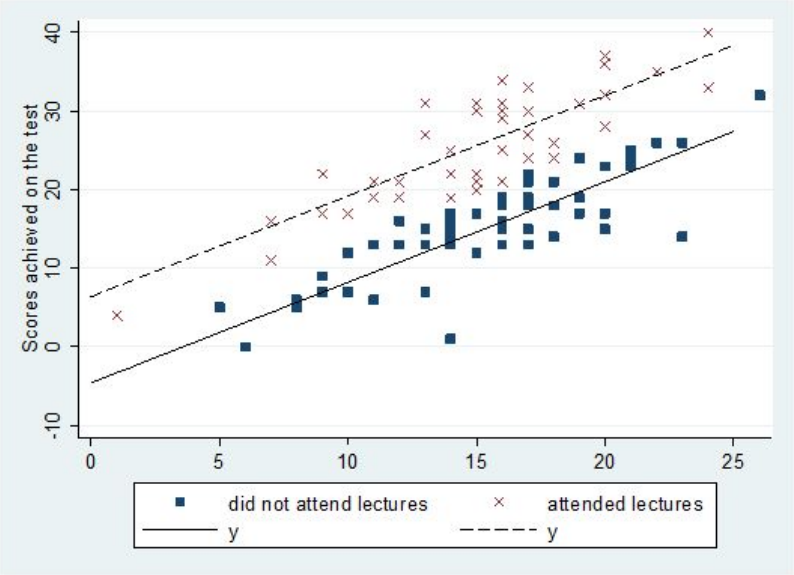
- Plot Y vs X, color/facet by Z
- Interaction signs
  - Different slopes by group ⟶ interaction candidate
  - Lines cross/diverge ⟶ strong interaction
- Parallel trends ⟶ interaction likely unnecessary

# Objective 3: Identify when interaction terms may be needed (2)

- Fit model without interaction: Y ~ X + Z
- Plot residuals vs X, separated by Z
- Different residual trends by Z
  - Add X:Z interaction and re-check

# Other Considerations:
# Factor (Categorical) Predictors – Main Effects

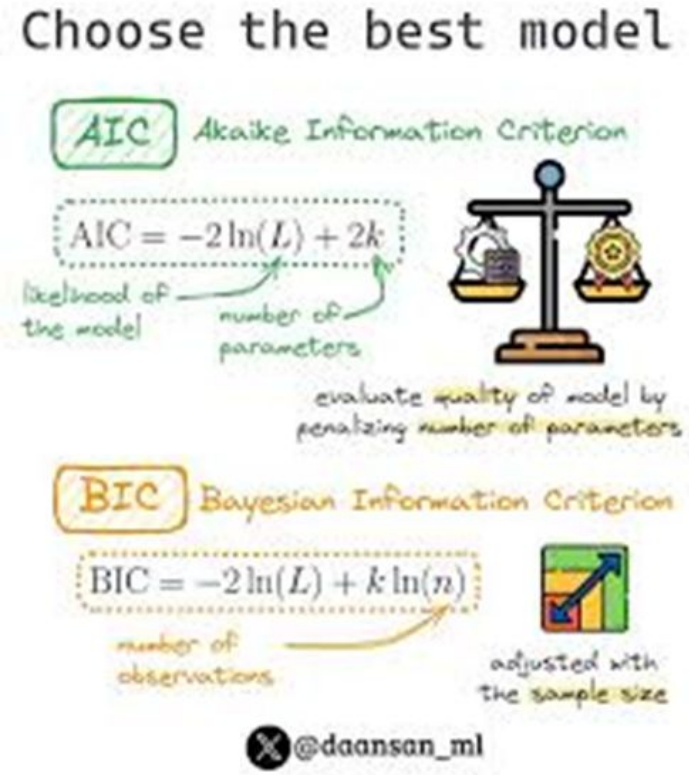
- Create dummy variable, X
  - X = 1, if the condition is met
  - X= 0, if the condition is not met
  - Example: X = 1, if attended lectures
    X = 0, if did not attend lectures




- Same slope/different intercept.
  - Inference on intercept indicates difference between the groups
- Multiple levels of the predictor: Create one dummy variable for each level
  - Example: Region with levels South, East, West, and North.
    *X1 = 1 if South, 0 if not X2 =1 if East, 0 if not, X3 = 1 if West, 0 if not, X4 = 1 if North, 0 if not*
- Model implicitly assumes that other predictors are unaffected by the factor predictor.

# Objective 4: Use AIC/BIC to evaluate whether diagnostic-driven changes improve the model

- Use diagnostics to propose a small set of model changes
- Compare baseline vs modified model (AIC/BIC)
- Keep change only if
  - AIC/BIC improves AND diagnostics improve
- Avoid blind searching across many terms
- **Model sequence example**
  - Base model: $Y \sim X$
  - Add polynomial: $Y \sim X + X^2$
  - Add interaction: $Y \sim X \times Z$
  - Compare AIC/BIC



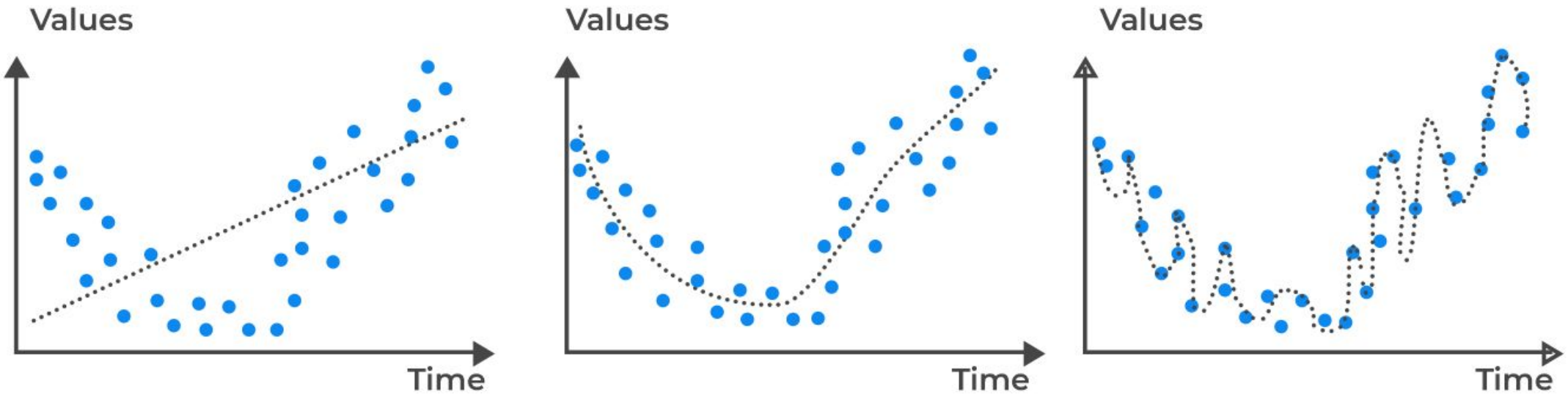
# Feature Engineering: Diagnostic and Transformation Summary

| Data Symptom (EDA/Residuals) | Visual Indicator | Recommended Transformation | Mathematical Goal |
|---|---|---|---|
| **Right Skew** | Histogram: Long tail to the right. | **Logarithm** log(x) or **Root** sqrt{x} | Pulls extreme outliers in; stabilizes variance. |
| **Left Skew** | Histogram: Long tail to the left. | **Square** x^2 or **Exp** e^x | Expands the higher end of the scale. |
| **Non-Linearity** | Residuals: U-shape or Inverted U-shape. | **Polynomial** x^2, x^3 | Captures the "bend" in the relationship. |
| **Heteroscedasticity** | Residuals: "Fan" or "Cone" shape. | **Log Target** log(y) or **Box-Cox** | Fixes non-constant variance in errors. |
| **Interaction** | Scatterplot: Slopes of y on x_1 change for different levels of x_2. | **Interaction Term** x_1 times x_2 | Accounts for variables that "depend" on each other. |
| **Categorical Impact** | Boxplot: Different medians/spreads across groups. | **Dummy Encoding** (0/1) | Converts qualitative labels into math. |

# Common Pitfalls

- **Chasing Curvature**
  - Slight bends due to noise
  - Overfitting polynomials
- **Spurious Interactions**
  - Small subgroup artifacts
  - Simpson’s paradox
- **Transformation Overuse**
  - Loss of interpretability
  - Domain meaning distortion

# Generalization and Overfitting




Underfitted

(High bias error)

Good Fit/Robust

(Balance between bias and variance)

Overfitted

( High variance error)

# Model Form Checklist

- Plot Y vs. X
- Fit simple linear model
- Examine residuals
- Ask:
  - Is the relationship curved?
  - Does variance change?
  - Do groups behave differently?
- Propose transformation or interaction
- Compare models using AIC/BIC
- Justify final model in words

# Part 2: Feature Inclusion Decisions (Importance/Selection)

# Why feature selection matters?

- Interpretability: fewer, meaningful coefficients
- Stability: correlated predictors $\longrightarrow$ unstable coefficients/SEs
- Generalization: too many predictors can overfit
- Communication: simpler model is easier to explain and defend

# Part 2 Main Objectives

- Distinguish model form (Part 1) vs. which predictors to include
- Use adjusted $R^2$ and AIC/BIC to compare predictor sets (linear regression)
- Use nested-model logic (partial F-test) to justify adding/removing predictors
- Diagnose multicollinearity (correlations, VIF) and choose stable predictors
- Combine statistics with domain reasoning to finalize a defensible model

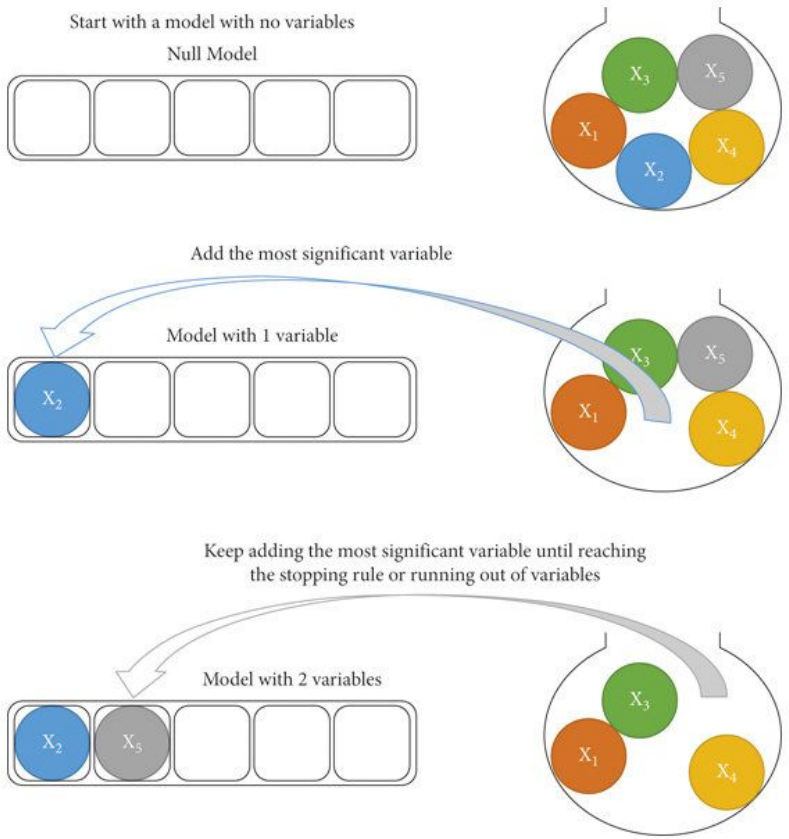
# Objective 1: Distinguish model form (Part 1) vs. which predictors to include (Part 2)

| Diagnostics | Feature Selection |
|---|---|
| Focus on **model form** | Focus on **predictors** |
| Identify curvature | Identify relevance |
| Identify interactions | Identify redundancy |
| Use residual plots | Use variance, correlation, tests |
| Answer *how* to model | Answer *what* to include |

# Which Variables are Important?
# Pre-Model Filtering for Candidate Predictors

- Variance Filtering
  - Constant predictors cannot explain outcomes
  - Near-constant predictors are often noise
  - This step is model-agnostic
- Correlation filtering
  - Highly correlated predictors distort interpretation
  - Diagnostics may look fine but coefficients become unstable
- Univariate screening
  - t-tests, simple regressions
  - Helps prioritize predictors
  - Still ignores interactions (which diagnostics later uncover)

- # Which are the Important Variables??



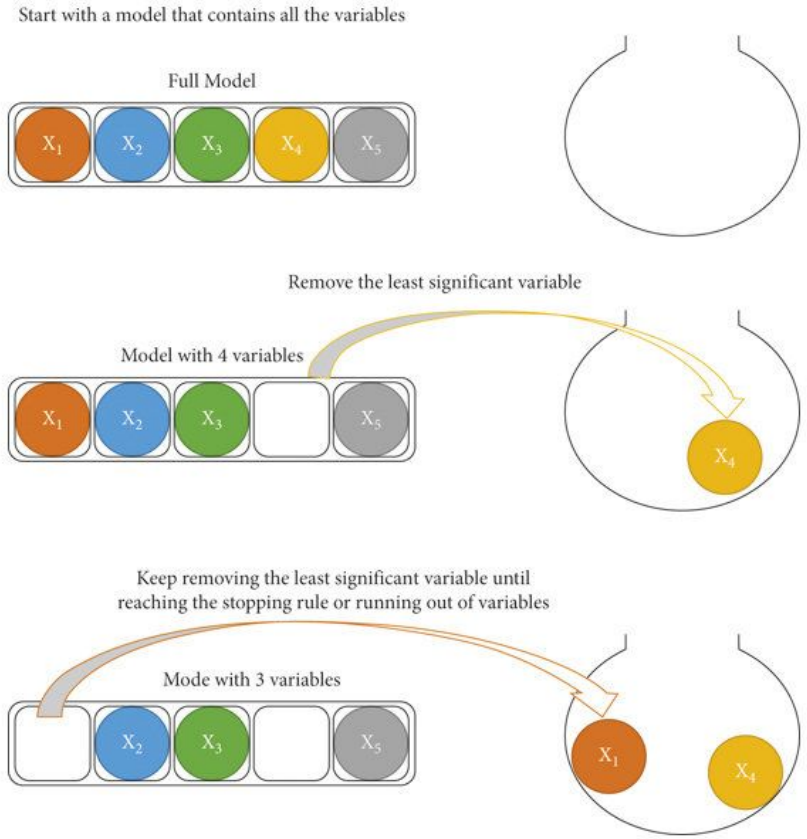




Image source: Computational Intelligence and Neuroscience, 2022(5):1-15

# Objective 2 - Use adjusted $R^2$ and AIC/BIC to compare predictor sets (backward stepwise selection) - 1

- Backward stepwise regression starts **fully loaded** and removes predictors one by one.
- It's most efficient when you have **many predictors** and enough data (since you fit fewer models than forward selection).
- Be cautious of **multicollinearity** — it can affect which variables appear "insignificant."
- Use **AIC**, **BIC**, or **adjusted $R^2$** alongside p-values for a balanced model selection approach.

## 📘 Backward Stepwise Regression: Summary Table

| Step | Action | Purpose | Key Decision Rule |
|---|---|---|---|
| 1️⃣ **Start with all predictors** | Fit a model that includes every candidate variable. | Begin with the most complex possible model. | — |
| 2️⃣ **Test all variables for significance** | Examine each predictor's t-statistic or p-value. | Identify which variable contributes the least to the model. | Choose the variable with the **highest p-value** (least significant). |
| 3️⃣ **Remove the least significant variable** | Drop the variable that provides the smallest contribution to explaining the outcome. | Simplify the model step by step. | Remove if its **p-value exceeds the removal threshold** (commonly > 0.10). |
| 4️⃣ **Refit the model** | Recalculate regression using the reduced set of predictors. | Update coefficient estimates and model fit statistics. | — |
| 5️⃣ **Reassess remaining variables** | Check remaining predictors again for significance. | Ensure all variables still meaningfully contribute. | Continue removing the variable with the next-highest p-value. |
| 6️⃣ **Repeat steps 4–5** | Continue eliminating one variable at a time. | Iteratively move toward a simpler, better-fitting model. | Stop when **all remaining predictors are statistically significant** ($p < 0.05$ or criterion met). |
| 7️⃣ **Final model selection** | Review the final reduced model. | Confirm it balances simplicity and explanatory power. | Evaluate using $R^2$, adjusted $R^2$, AIC/BIC, and diagnostic plots. |

↓

# Objective 2 - Use adjusted $R^2$ and AIC/BIC to compare predictor sets (forward stepwise selection) - 2

- Forward stepwise regression is a **greedy algorithm** — it picks the "best" variable at each step, but may not find the globally best combination.
- Common **selection criteria**:
  - **Statistical significance** ($p < 0.05$)
  - **Information criteria** (lowest AIC or BIC)
  - **Adjusted $R^2$** (largest increase)
- Always run **diagnostics** on the final model to check assumptions and multicollinearity.

# Objective 2 - Use adjusted $R^2$ and AIC/BIC to compare predictor sets bidirectional stepwise selection) - 3

- **Bidirectional selection** combines both forward and backward approaches — adding and removing predictors at each stage.
- It's more flexible and often finds a **better balance** between underfitting and overfitting.
- Requires careful choice of **entry and exit criteria** (e.g., p-in < 0.05, p-out > 0.10).
- Commonly guided by **AIC/BIC minimization** rather than strict p-values.
- Always validate the final model using **cross-validation** and **residual diagnostics**

## 📘 Bidirectional (Both) Stepwise Regression: Summary Table

| Step | Action | Purpose | Key Decision Rule |
|---|---|---|---|
| 1️⃣ **Start with an initial model** | Begin with no predictors, or a small starting model (often the intercept only). | Establish a baseline model to expand and refine. | — |
| 2️⃣ **Forward step:** Test all variables not yet in the model. | Add the variable that most improves model fit (largest $R^2$ increase or smallest AIC/BIC). | Gradually build the model by including significant predictors. | Add if **p-value < inclusion threshold** (e.g., 0.05) or **reduces AIC/BIC**. |
| 3️⃣ **Add the best variable** | Include the variable that passes the inclusion rule. | Expand the model while monitoring fit. | — |
| 4️⃣ **Backward step:** Re-test all variables already in the model. | Check if any variables have become non-significant after the new addition. | Keep the model parsimonious (avoid unnecessary predictors). | Remove any variable with **p-value > exclusion threshold** (e.g., 0.10). |
| 5️⃣ **Refit the model** | Recalculate the regression after any additions or deletions. | Update coefficients and test statistics. | — |
| 6️⃣ **Repeat forward and backward steps** | Continue alternating between adding and removing predictors. | Allow the model to correct earlier inclusion or exclusion mistakes. | Stop when no variable can be added or removed based on thresholds. |
| 7️⃣ **Final model selection** | Review the final s predictors. | Ensure the model balances fit, simplicity, and interpretability. | Evaluate $R^2$, adjusted $R^2$, AIC/BIC, and run regression diagnostics. |

# Objective 3: Use nested-model logic (partial F-test) to justify adding/removing predictors

- *Question: Can we use a simpler linear model nested within a complex one?*
- Compare a smaller model vs a larger model that contains it
- Ask: does the added predictor(s) reduce error more than we'd expect by chance?
- Use when models are nested and assumptions are reasonable
- Communicate results in plain language (not just p-values)

# Pseudo-code Example

Input: x (independent variable), y (dependent variable)

Output: Parameters of both models, p-value, and decision

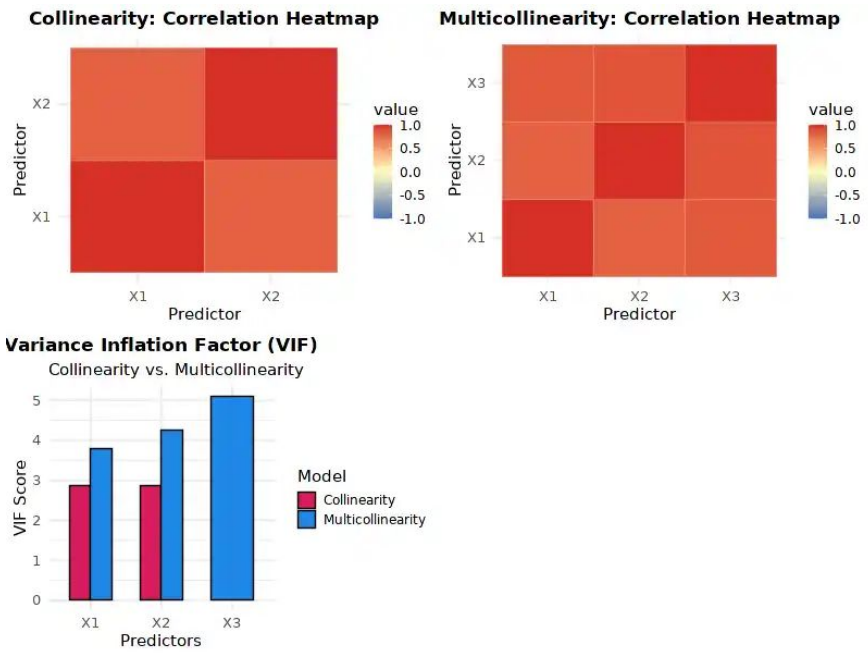
```
1. Define Simple Model: y_simple = a*x + h
2. Define Complex Model: y_complex = a*x + h + c*x / (d + x)
3. Fit the Simple Model to the data (x, y) to find parameters and compute RSS1.
4. Fit the Complex Model to the data (x, y) to find parameters and compute RSS2.
5. Compute F-statistic:
    F = ((RSS1 - RSS2) / (p2 - p1)) / (RSS2 / (n - p2))
6. Compute p-value from the F-distribution with (p2 - p1, n - p2) degrees of freedom.
7. If p-value < alpha (e.g., 0.05):
     Print "The complex model is significantly better."
  Else:
     Print "The simpler model is sufficient."
8. Plot the data with both model fits for visualization (optional).
```

# Objective 4: Diagnose multicollinearity (correlations, VIF) and choose stable predictors

- Correlated predictors make coefficients unstable
- Standard errors inflate → p-values become unreliable
- Coefficient signs can flip across samples
- Prediction may be fine, interpretation suffers

**Strategies:**

- Remove Highly Correlated Features:
  - Drop one of the correlated features, the one which is less important or with a higher VIF. Removing such features reduces redundancy and improves model interpretability and stability.
- Combine Variables or Use Dimensionality Reduction Techniques
  - Create new variables such as BMI from height and weight.
  - Use Principal Component Analysis (PCA) to convert correlated




$$VIF_i = \frac{1}{1 - R_i^2}$$

Image Source: https://metricgate.com/blogs/collinearity-vs-multicollinearity/

# Objective 5: Combine statistics with domain reasoning to finalize a defensible model

- Remove identifiers and leakage variables
- Prefer predictors with clear meaning and availability at prediction time
- Keep policy/science-relevant variables even if marginally weaker
- Document decisions: why kept/dropped

# Part 3: Intro to Model Evaluation

# Why Evaluate Model (Predictive) Performance?

- Is the model accurate enough to make reliable predictions?
- Does it generalize well to unsee data?
- How does it compare to other models or benchmarks?
- What are the limitations and weaknesses of the model?

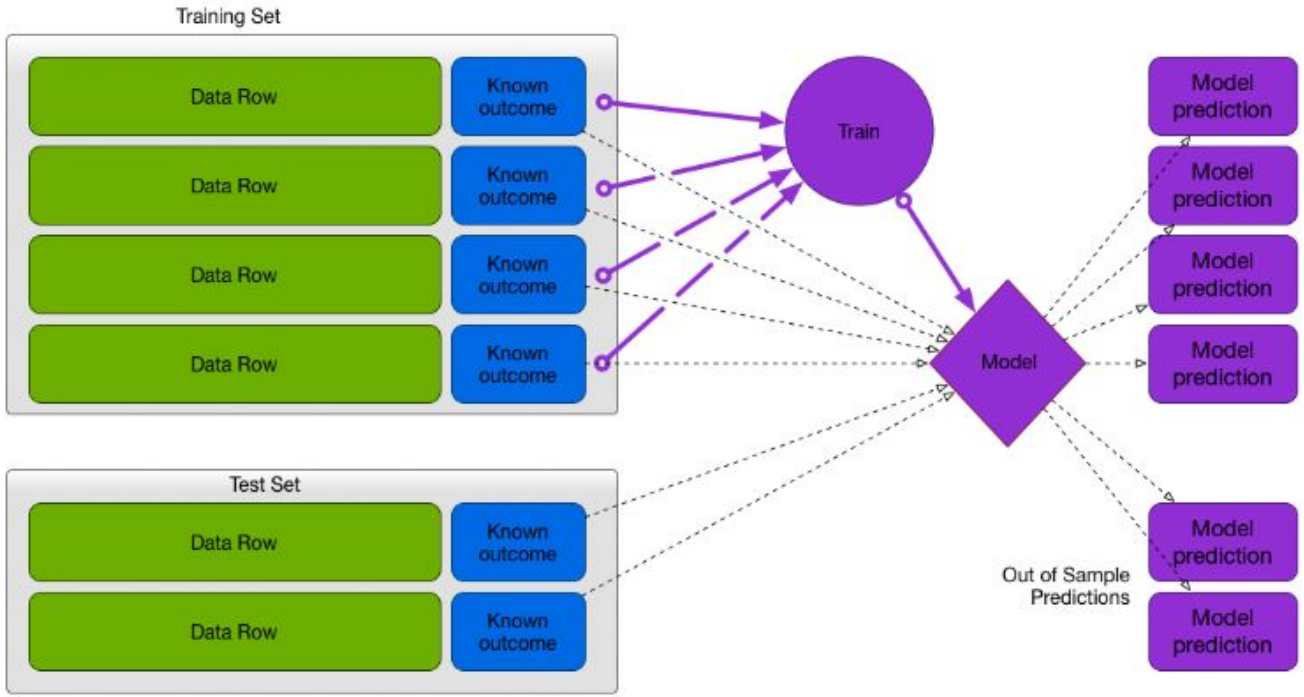
"Essentially, all models are wrong, but some are useful"

George E.P. Box

# Common Metrics for Performance Evaluation

- MSE (Mean Squared Error)
  - Advantages: differentiable, so easy to use as loss function
  - Disadvantages: squared unit output (interpretation); not robust to outliers
- RMSE (Root Mean Squared Error)
  - Advantages: same unit as output
  - Not robust to outliers
- MAE (Mean Absolute Error)
  - Advantages: Same unit as output variable; most robust to outliers
  - Disadvantages: Not differentiable (for optimization), so need optimizers like Gradient descent

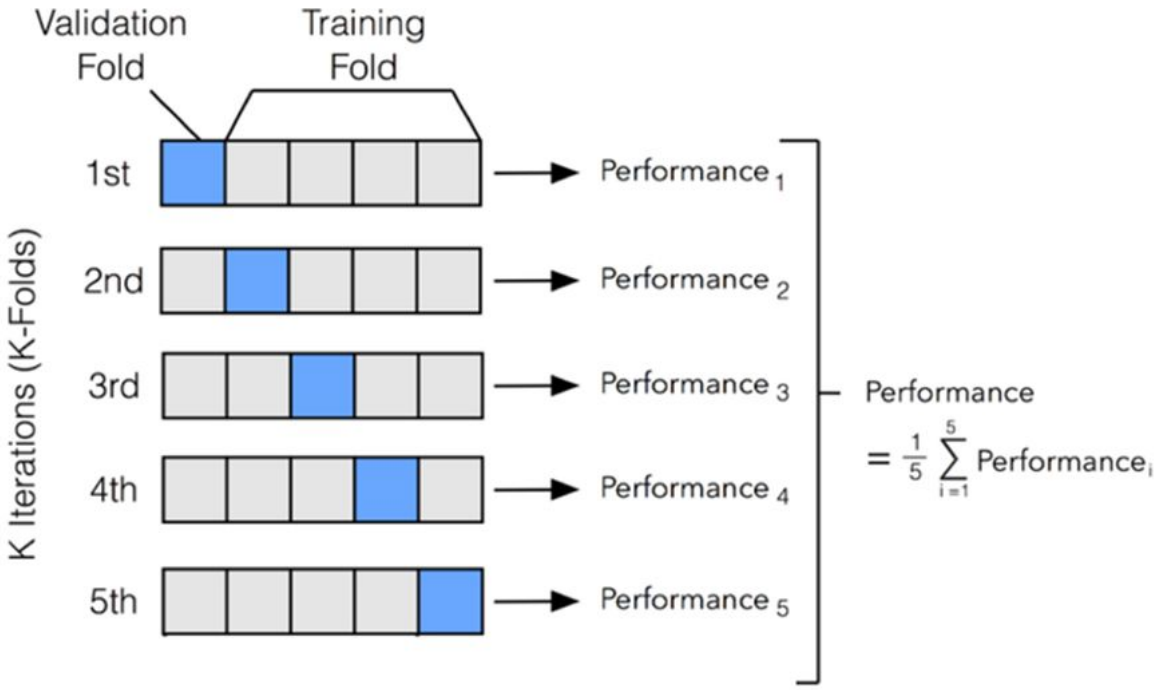
# Train/Test Split



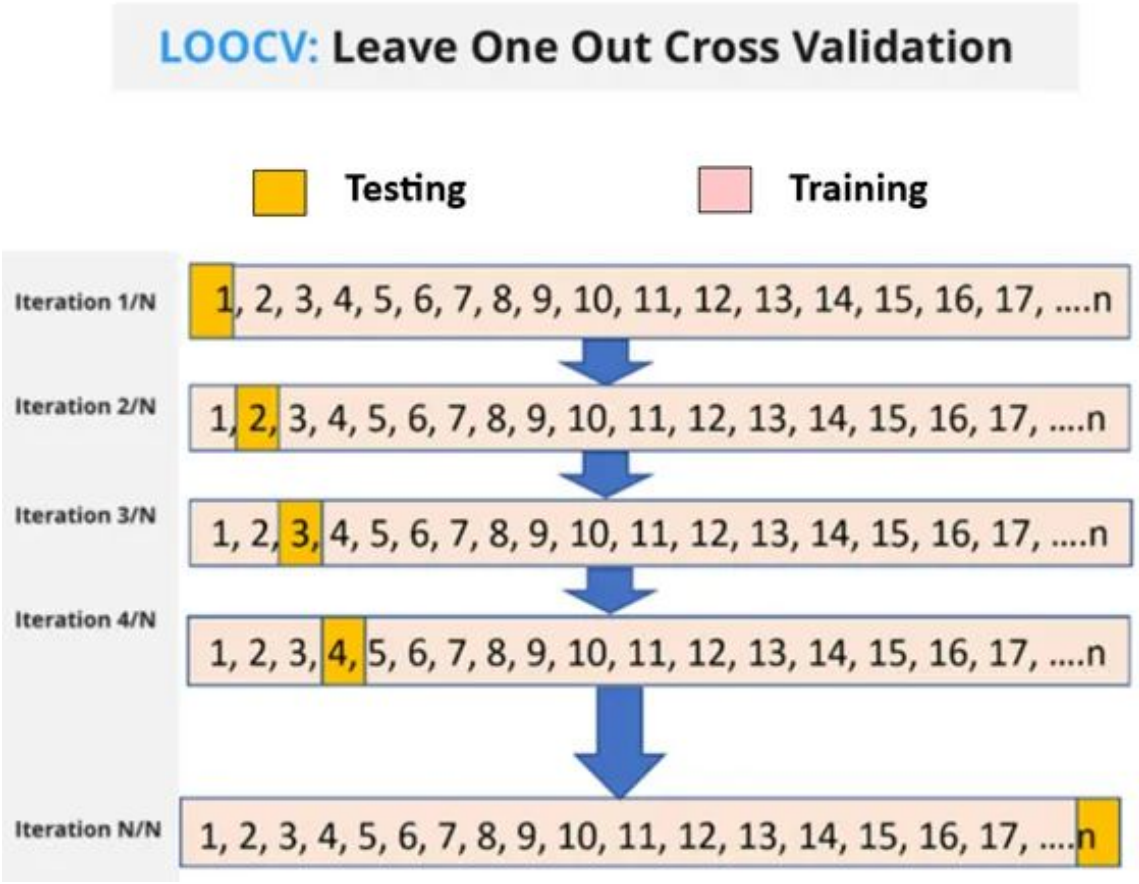

Recommended method when data is plentiful

# K-fold Cross-Validation

- Split data into K subsets or "folds" of equal size
- Train on K-1 folds and test on the remaining fold
- Repeat K times, each fold being used once as test
- Average performance across K iterations

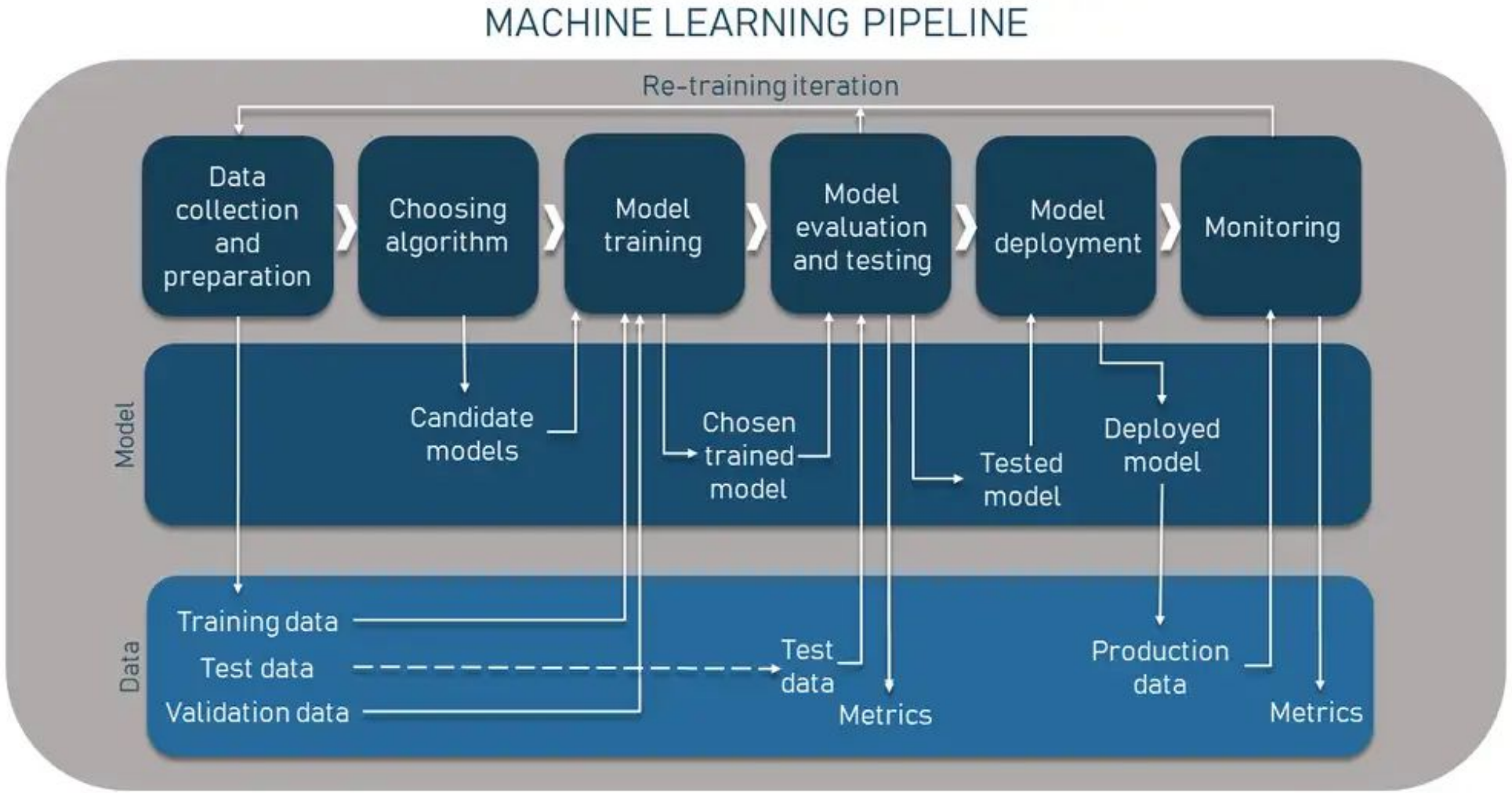


# LOOCV: Leave One Out Cross Validation




- Select n samples of the dataset
- Use n-1 samples of the dataset as the training set and 1 sample as the test set
- Repeat n times, each time leaving out a different sample as the test set
- Known as an exhaustive technique
- Pro:  high accuracy (uses most of the samples of training set)
- Con: computationally expensive for large data

# A More Detailed/Formal Pipeline




altexsoft